To the memory of Raien

An Introduction to Topological Graph Theory

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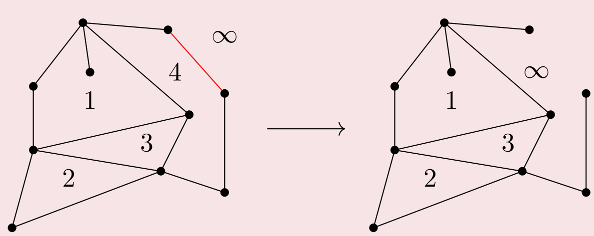
Abstract

If you have ever studied graph theory, you have most likely encountered Euler's well-known formula for planar graphs. This theorem states that given a finite, connected planar graph (a graph drawn in such a way that no two distinct edges intersect) in the plane, then the equality $|V| - |E| + |F| = 2$ holds, where $|V|$ denotes the number of vertices, $|E|$ the number of edges, and $|F|$ the number of regions into which the plane is divided by the graph. (precise definition of F requires preliminary notions that will be introduced later.) Since this relation does not hold for a very large class of graphs, it is natural to ask whether the number 2 is related to the planarity of the graph, and more importantly, in other cases, what the quantity $|V| - |E| + |F|$ reflects about the graph.

Until precise mathematical definitions of the notions of "the number of regions" and "planar graph" are introduced, rely on their intuitive meanings: A planar graph is a graph that "can be drawn" in the plane in such a way that any two distinct edges intersect only at their common vertices. Note that it is more accurate to speak of a "planar sketch of a graph" rather than a "planar graph", since a planar graph may clearly be drawn in the plane in such a way that distinct edges intersect. In fact, a planar graph is precisely a graph that admits a planar sketch. It is appropriate to begin by proving some elementary results concerning Euler's formula:

Theorem 1. Given a planar sketch of a finite connected planar graph, one has $|V| - |E| + |F| = 2$.

Proof. Note that the unbounded region of the plane is also counted as one of the regions of the graph, and therefore $|F|$ is always a positive integer. We prove the statement by induction on $|F|$. For the base case, assume that $|F| = 1$. Observe that such a graph cannot contain a cycle. (indeed, the portion of the planar sketch corresponding to a cycle partitions the plane into exactly two connected components, and hence the graph partitions the plane into at least two connected components. A rigorous proof of this claim requires the Jordan curve theorem.) Since our graph is connected and acyclic, it must be a tree, and by a basic result from graph theory we know that for a tree, $|E| = |V| - 1$. Hence $|V| - |E| + |F| = |V| - (|V| - 1) + 1 = 2$, establishing the base case of the induction. Now, for the induction step, assume that the statement holds for every graph whose number of regions is less than n. Consider a graph with $|F| = n + 1$. Since $|F| \geq 2$, there exists a region of the graph whose boundary shares an edge with the boundary of the unbounded region. First observe that removing this edge preserves the connectedness of the graph. (since this edge lies on the boundary of a region bounded by a cycle, it is not a cut-edge.) Next, note that removing this edge decreases both $|E|$ and $|F|$ by exactly one. Therefore, by the induction hypothesis, for graphs with n regions we have $|V| - |E| + |F| = |V| - (|E| - 1) + (|F| - 1) = 2$, which completes the proof. $\square$



An example of the inductive process of the proof

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Corollary 1. The value $|F|$ is the same for every planar sketch of a graph and is therefore an invariant of G.

Corollary 2. Given a finite connected planar graph with at least 3 vertices, one has $|E| \leq 3|V| - 6$.

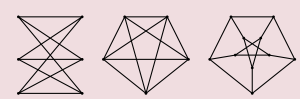
Proof. In every graph other than the path of length 2, the number of vertices (and consequently the number of edges) on the boundary of each region is at least 3. Therefore, the degree of each region (or equivalently, the degree of the corresponding vertex in the dual graph) is at least 3, and hence the sum of the degrees of the vertices of the dual graph is at least $3|F|$. On the other hand, each edge is either incident with the boundaries of two adjacent regions or lies only on the boundary of a single region (such as the edge contained in region 1 in the proof of theorem 1). In the former case, this edge is counted once in the degree of each of the two adjacent regions, while in the latter case it gives rise to a loop at the vertex corresponding to that region in the dual graph. Therefore, each edge contributes exactly 2 to the sum of the degrees of the vertices of the dual graph, and thus this sum is equal to $2|E|$. Combining, we obtain $2|E| \geq 3|F|$. Finally, Euler's formula yields:

$$2 = |V| - |E| + |F| \leq |V| - |E| + \frac{2}{3}|E| = |V| - \frac{1}{3}|E| \implies 6 \leq 3|V| - |E| \implies |E| \leq 3|V| - 6 \qquad \square$$

Corollary 3. Given a finite planar graph with at least 3 vertices and no triangles, one has $|E| \leq 2|V| - 4$.

Proof. Assume first that the given graph is connected. Observe that, by the assumption of the problem, the boundary of each region contains at least 4 edges. Therefore, proceeding exactly as in the previous proof, we obtain $2|E| \geq 4|F| \iff |E| \geq 2|F|$, and consequently the desired inequality follows from Euler's formula in the same manner. Now prove the statement for an arbitrary graph. $\square$

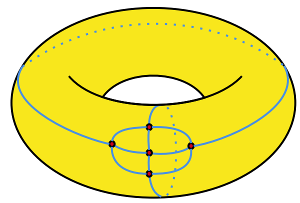
Problem 1. Prove that the Petersen graph, K_5, and $K_{3,3}$ are not planar.



Having established these elementary results, we can now give a more precise meaning to the notions of a "planar graph" and "regions determined by a planar graph". As mentioned earlier, a graph is planar precisely when it admits a planar sketch. Therefore, we must first define precisely what is meant by a planar sketch of a graph.

Definition 1. Suppose that a sketch of a graph in the plane is regarded as a map $\rho : G \to \mathbb{R}^2$ that sends each vertex $v \in V(G)$ to a point $\rho(v) \in \mathbb{R}^2$ and each edge $\overline{uv} = e \in E(G)$ to a path between $\rho(u)$ and $\rho(v)$ in the plane, such that $\rho\big|_V$ is injective and, moreover, for every two distinct edges $e_1, e_2 \in E(G)$, the images of these edges $(\rho(e_1), \rho(e_2) \in \mathbb{R}^2)$ are disjoint subsets, except when both e_1 and e_2 are incident with a vertex $v \in V(G)$, in which case $\rho(e_1) \cap \rho(e_2) = \{\rho(v)\}$. Such a map ρ is called a graph embedding of G in the plane.

You (hopefully) proved in problem 1 that the graph K_5 is not planar, or equivalently, that there is no embedding of K_5 in the plane. Now consider the following figure:



As can be seen, the graph K_5 is embedded on the surface of a torus (donut) without any additional intersections. Therefore, if our criterion for the planarity of a graph were the existence of an embedding on the torus, then K_5 would also be a planar graph. On the other hand, one can see that this graph determines 5 regions on the torus. Consequently, computing $|V| - |E| + |F|$ yields the value zero. To understand the meaning behind this observation, we first require several tools from point-set topology.

Problem 2. A finite graph admits a planar embedding if and only if it admits an embedding on a sphere.

Problem 3. Prove that every planar graph G admits an embedding on the torus.

Definition 2. For every $x_0 \in \mathbb{R}^3$ and $r > 0$, the open ball of radius r centered at x_0 is denoted by $B(x, r)$. Thus, $B(x, r) = \{x \in \mathbb{R}^3 \mid |x - x_0| < r\}$. A subset $A \subseteq \mathbb{R}^3$ is called open if, for every $x \in A$, there exists a sufficiently small $\epsilon > 0$ such that $B(x, \epsilon) \subseteq A$. The complement of an open subset is called a closed subset.

Example 1. For every $x \in \mathbb{R}^3$ and $r > 0$, the open ball $B(x, r)$ is an open subset of $\mathbb{R}^3$.

Example 2. The set of points $\{(x, y, z) \in \mathbb{R}^3 \mid x, y, z \in \mathbb{Z}\}$ form a closed subset of $\mathbb{R}^3$.

Example 3. The set of points $\{(x, y, z) \in \mathbb{R}^3 \mid x, y, z \in \mathbb{Q}\}$ is neither open nor closed.

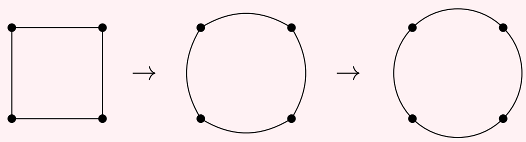
Problem 4. Prove that a subset $A \subseteq \mathbb{R}^3$ is closed if and only if it contains all of its limit points. This means that if we consider a convergent sequence $\{x_n\}_{n=1}^{\infty} \to x$ such that $\forall n \in \mathbb{N} : x \neq x_n \in A$, then $x \in A$. For example, consider the sequence of points $x_n = (0, 0, \frac{n-1}{n})$ in the open unit ball centered at the origin. This sequence converges to the point $x = (0, 0, 1)$, which, by definition, does not belong to this open ball. Hence, it follows that this open ball (and similarly, every other open ball) is not a closed subset of $\mathbb{R}^3$.

Definition 3. Suppose that open subsets $A, B \subseteq \mathbb{R}^3$ and a function $f : A \to B$ are given. Then f is said to be continuous at $x \in A$ if, for every open ball $B(f(x), \epsilon)$ centered at $f(x)$, there exists an open ball $B(x, \delta)$ centered at x such that $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$. By the definition of an open ball, this condition may be written as $\forall \epsilon > 0 : [\exists \delta > 0 : \forall x' \in \mathbb{R}^3 : (|x - x'| < \delta \implies |f(x) - f(x')| < \epsilon)]$. We furthermore say that f is continuous if it is continuous at every $x \in A$.

Definition 4. Suppose that open subsets $A, B \subseteq \mathbb{R}^3$ and a continuous bijection $f : A \to B$ are given such that the inverse of f (viewed as a function $f^{-1} : B \to A$) is also continuous. Then f is called a homeomorphism, and the sets A and B are said to be homeomorphic. We write $A \sim B$.

Example 4. Prove that any two open balls in $\mathbb{R}^3$ are homeomorphic. Prove that any two similar open sets are homeomorphic. Also prove that any two line segments in $\mathbb{R}^2$ are homeomorphic.

Example 5. Prove that the open square defined by $\{(x, y) \in \mathbb{R}^2 \mid |x| < 1, |y| < 1\}$ is homeomorphic to the open disk $\{(x, y) \in \mathbb{R}^2 \mid \sqrt{x^2 + y^2} < 1\}$. To what extent can you generalize this example?



Problem 5. The notion of homeomorphism in topology provides a meaning of equality for two subsets of a space. To show that this notion indeed possesses the properties of equality, prove that homeomorphism induces an equivalence relation on the collection of all subsets of $\mathbb{R}^3$.

Reminder. A relation $\sim$ on the elements of a set S is called an equivalence relation if the following hold:

1. $\forall a \in S : a \sim a$
2. $\forall a, b \in S : a \sim b \iff b \sim a$
3. $\forall a, b, c \in S : (a \sim b, b \sim c) \implies a \sim c$

Problem 6. Prove that an equivalence relation on S partitions the set S into equivalence classes consisting of equivalent elements. By the equivalence class of $a \in S$ we mean $[a] = \{b \in S \mid a \sim b\}$.

Consequently, since homeomorphism defines an equivalence relation on the subsets of $\mathbb{R}^3$ (and more generally of every $\mathbb{R}^n$), one may consider the homeomorphism classes of subsets of $\mathbb{R}^3$. From the point of view of topology, two homeomorphic subsets are indistinguishable. In other words, the homeomorphism relation serves as a notion of equality between two topological spaces, and every topological property is preserved under a homeomorphism.

Remark 1. These definitions can be generalized to any $\mathbb{R}^n$. It is worthwhile to formulate them for $n = 2$.

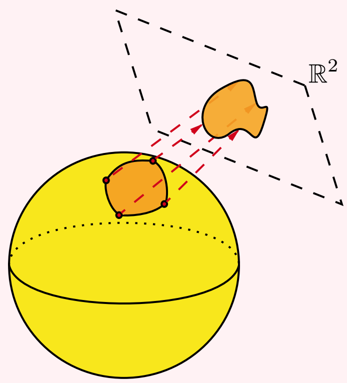
Definition 5. A path in A between $x, y \in A \subseteq \mathbb{R}^3$ is a homeomorphism $f : [0, 1] \to A$ where $f(0) = x, f(1) = y$.

Problem 7. Define the relation $\to$ on the points of a subset $A \subseteq \mathbb{R}^3$ by declaring that $\forall x, y \in A : x \to y$ if there exists a path from x to y in A. Prove that this relation is an equivalence relation on the points of A, and therefore partitions A into equivalence classes. The equivalence classes of this relation are called the connected components of A. If A has only one connected component, then it is called connected.

We can now finally give a precise definition of the "regions of an embedding". By the regions of an embedding $\rho : G \to \mathbb{R}^2$ we mean the connected components of the set $\mathbb{R}^2 \setminus \rho(G)$, which we denote by F. Consequently, $|F|$ is the number of connected components. Observe that for an embedding of a finite graph in the plane, all connected components of $\mathbb{R}^2 \setminus \rho(G)$ except for one are bounded, while the last connected component is unbounded; this is called the unbounded region. Note that this definition depends crucially on the embedding map ρ. Also observe that, for example, in an embedding of a graph on the sphere, there is no unbounded region, and all regions of the embedding are bounded (since the sphere itself is a bounded set).

Problem 8. Continuing problem 2, prove that the number of regions of an embedding of a graph in the plane is equal to the number of regions of an embedding of the graph on the sphere.

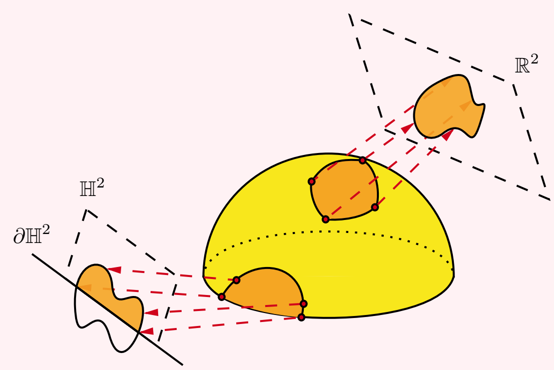
Definition 6. A subset $M \subseteq \mathbb{R}^3$ is called a surface without boundary if it is locally homeomorphic to the plane. This means that for every $x \in M$ and every arbitrarily small open ball $B(x, \epsilon)$ centered at x, the neighborhood $M \cap B(x, \epsilon)$ is homeomorphic to an open subset $U \subseteq \mathbb{R}^2$. A surface need not itself be homeomorphic to the plane. For example, the sphere is not homeomorphic to the plane, but it is locally homeomorphic to the plane.



With a little care, one realizes that many objects which at first glance appear to be surfaces are in fact not surfaces. For example, the closed unit disk ($x^2 + y^2 \leq 1$) might seem to be a surface in $\mathbb{R}^3$, but upon closer inspection one sees that for a point on the boundary of the disk, there is no neighborhood of that point in the closed unit disk that is homeomorphic to an open subset of $\mathbb{R}^2$. In a sense, it is the boundary of the disk that prevents it from being a surface. On the other hand, surfaces naturally acquire boundaries after operations such as "cutting." Consequently, the definition of a surface must be generalized so as to accommodate the notion of a "boundary".

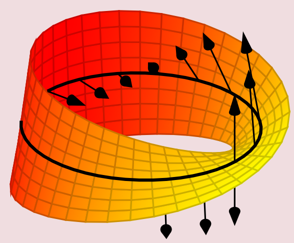
Definition 7. By the nonnegative half-plane we mean the subset $\mathbb{H}^2 := \{(x, y) \in \mathbb{R}^2 \mid y \geq 0\} \subsetneq \mathbb{R}^2$. A subset of $\mathbb{H}^2$ is called open if it is the intersection of an open subset of $\mathbb{R}^2$ with $\mathbb{H}^2$ (for example, the upper open half-disk $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\} \cap \{(x, y) \in \mathbb{R}^2 \mid y \geq 0\}$ is an open subset of $\mathbb{H}^2$).

Definition 8. A subset $M \subseteq \mathbb{R}^3$ is called a surface with boundary if there exists a subset $\partial M \subseteq M$ such that for every $x \in M \setminus \partial M$ and every arbitrarily small open ball $B(x, \epsilon)$ centered at x, the neighborhood $M \cap B(x, \epsilon)$ is homeomorphic to an open subset $U \subseteq \mathbb{R}^2$, and for every $x \in \partial M$ and every arbitrarily small open ball $B(x, \epsilon)$ centered at x, the neighborhood $M \cap B(x, \epsilon)$ is homeomorphic to an open subset of $\mathbb{H}^2$. In this case, ∂M is called the boundary of the surface M.



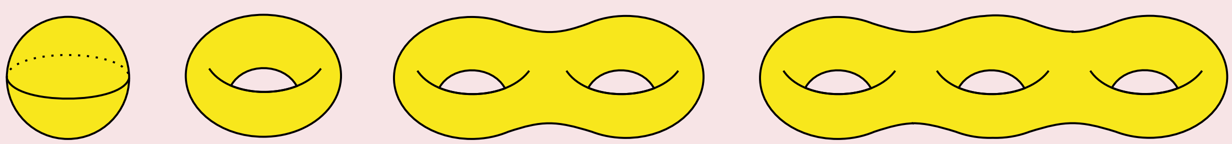
A precise definition of an orientable surface requires some background in differential geometry. Therefore, we shall for now settle for an intuitive definition. Since a surface is locally similar to either the plane or the half-plane at every point, each surface locally determines two "sides" of the ambient space at every point. If one can choose one of these two "sides" at every point of the surface by means of a vector in such a way that the resulting assignment of vectors to points of the surface is continuous, then the surface M is called orientable. The sphere is an example of an orientable surface, since one may clearly choose at every point the vector directed toward the center of the sphere, and this assignment is continuous. On the other hand, the Möbius band is an example of a non-orientable surface.

Problem 9. Using the definitions and the figure below, prove that the Möbius band is non-orientable.



We say that a surface M is closed if it is bounded and $M \subseteq \mathbb{R}^3$ is a closed subset (not to be confused with the notion of merely being a closed subset). We now state, without proof, a fundamental theorem known as the "classification theorem for orientable surfaces", which determines the homeomorphism classes of connected, closed, orientable surfaces.

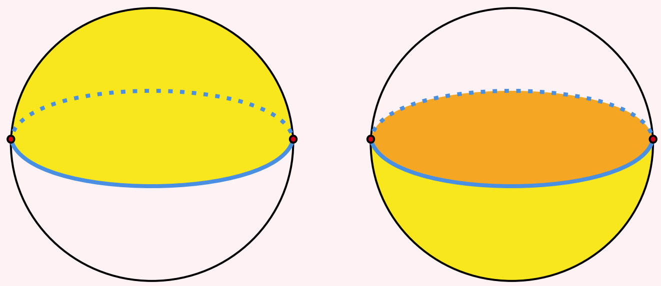
Theorem 2. Let M_g denote the surface of a torus with g holes. Then every connected, closed, orientable surface M is homeomorphic to exactly one of the surfaces M_g. The integer g is called the genus of the surface M and is denoted by $\gamma(M)$.



But how are graphs related to surfaces? To explain this, we first present a combinatorial way of defining surfaces. From now on, the term "surface" will refer to both surfaces with boundary and surfaces without boundary.

Definition 9. First consider the set X_0 of embeddings of finitely many points (the 0-dimensional closed ball) in $\mathbb{R}^3$. (The images of these embeddings form a set of points in space.) Next, consider the set X_1 of embeddings of finitely many unit line segments (the 1-dimensional closed ball) in $\mathbb{R}^3$ such that $\forall(\sigma : I \to \mathbb{R}^3) \in X_1 : \sigma\big|_{\partial I} \subseteq X_0$. (Here ∂I denotes the boundary of I, and hence $\partial I = \{0, 1\}$.) In other words, X_1 is obtained by attaching finitely many line segments between the points selected in X_0. Finally, consider the set X_2 of embeddings of finitely many unit disks (the 2-dimensional closed ball) in $\mathbb{R}^3$ such that $\forall(\sigma : \mathbb{D} \to \mathbb{R}^3) \in X_2 : \sigma\big|_{\partial\mathbb{D}} \subseteq X_1$. In other words, X_2 is obtained by attaching finitely many disks so that their boundaries lie in X_1. Then $M := \bigcup X_i \subseteq \mathbb{R}^3$ defines a surface. The set X_0 consisting of the embeddings of points in space is called the 0-skeleton of the surface M. Similarly, X_1 is called the 1-skeleton and X_2 the 2-skeleton of the surface M. Such a structure is called a CW-complex for the surface M. The CW-complex associated with a surface is not unique.

Example 6. A CW-complex of the sphere: the points of X_0 are shown in red, the edges of X_1 in blue, and the faces of X_2 in yellow. Justify that this structure defines a CW-complex.



Definition 10. For a surface M equipped with a CW-complex structure, the Euler characteristic of the surface is defined by $\chi(M) = |X_0| - |X_1| + |X_2|$. One can prove that homeomorphic spaces have the same Euler characteristic. Consequently, the Euler characteristic of a surface is independent of the CW-complex structure.

Both the genus and the Euler characteristic are invariants under homeomorphism. In fact, one can prove the following:

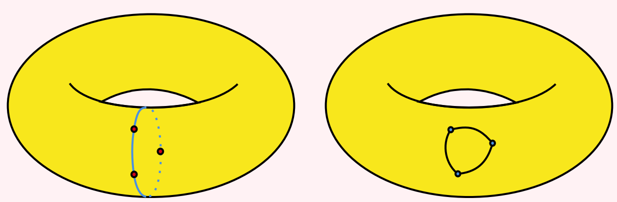
Theorem 3. For every connected, closed, orientable surface M, the relation $\chi(M) = 2 - 2\gamma(M)$ holds between the genus and the Euler characteristic. In other words, they are the same invariant.

Now let M be a connected surface equipped with a CW-complex structure. Observe that the 1-skeleton of this CW-complex is a connected graph on the surface M. (why?) Conversely, if an arbitrary graph is embedded on the surface M and all the regions determined by the graph are homeomorphic to the 2-dimensional closed ball (the disk), then one can easily construct a CW-complex structure on M from this graph.

Definition 11. Suppose that a connected closed surface M, a graph G, and an embedding $\rho : G \to M$ are given such that all the regions determined by G on the surface M are homeomorphic to the 2-dimensional closed ball (the disk). Then ρ is called a cellular embedding.

Remark 2. For a graph G and a surface M, the number of regions of G on M depends on the embedding map and is not uniquely determined by M and G.

Example 7. In the figure, the triangle graph is depicted with two different embeddings on the torus. In the first case (shown in red), the number of regions determined by the graph on the torus is 1, whereas in the second case (shown in blue), the number of these regions is 2.



The number of regions determined by G on M under the embedding ρ is denoted by $|F_\rho|$, and is equal to the number of connected components of $M \setminus \rho(G)$.

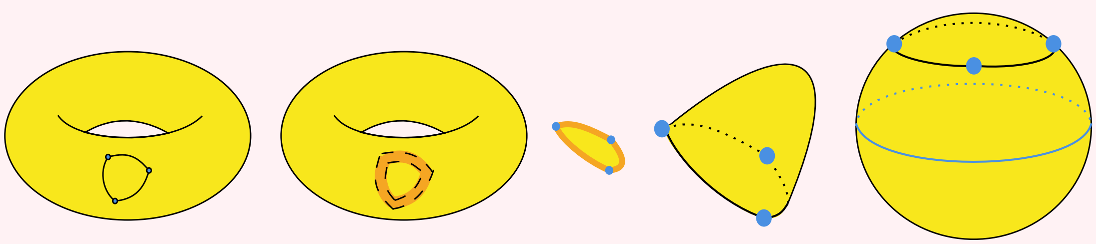
Consequently, we have in fact proved the following important theorem:

Theorem 4. Generalized Euler formula: For every finite connected graph G admitting a cellular embedding $\rho : G \to M$ on a connected closed surface M, the equality $|V| - |E| + |F| = \chi(M)$ holds. In the case $M = \mathbb{R}^2$, this theorem reduces to Euler's formula. (why?)

Corollary 4. Suppose that a finite connected graph G, an arbitrary connected closed surface M, and cellular embeddings $\rho_1, \rho_2 : G \to M$ are given. Then the numbers of regions determined by G on M under the embeddings ρ_1 and ρ_2 are equal. (in other words, $|F_{\rho_1}| = |F_{\rho_2}|$)

Thus, cellular embeddings form a well-behaved class of embeddings for studying the structure of a graph. However, as you saw in example 7, not every embedding is cellular. Nevertheless, one hopes to replace the connected components that are not homeomorphic to a disk by disks so that the resulting embedding of the graph on the new surface becomes cellular. The following construction allows us to transform a non-cellular embedding into a cellular one:

Definition 12. **The capping operation**[7]: Suppose that $\rho : G \to M$ is a non-cellular embedding of a finite connected graph on a connected closed surface. Let $S \subseteq M \setminus \rho(G)$ be a connected component that is not homeomorphic to a disk (the 2-dimensional closed ball). Denote the boundary of S by ∂S and define $\overline{S} = S \cup \partial S$. Provided that $M \setminus \overline{S}$ is a surface, we may remove S from the surface and attach a disk (the cap) so that the boundary of the disk coincides with ∂S. Consequently, this region becomes homeomorphic to a disk.



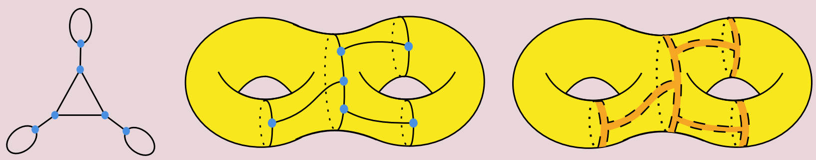
Remark 3. Contrary to what might appear at first glance, $M \setminus \overline{S}$ is not necessarily a surface. As an example, consider the other embedding from example 7. Identify the obstruction.

To overcome this difficulty, we state the following lemma without proof. The importance of this lemma lies in its geometric intuition, so it would be worthwhile to study its example thoroughly.

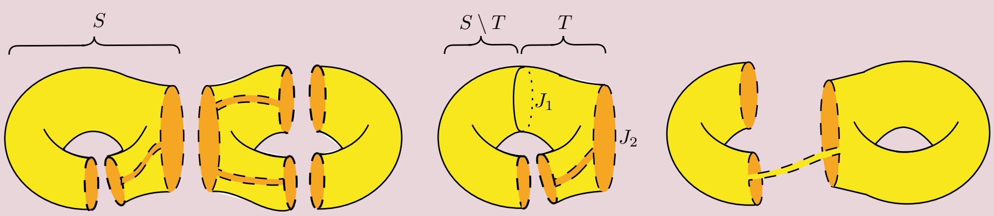
Lemma 1. Let M be a connected closed surface, and let $\varnothing \neq K \subsetneq M$ be a connected, closed, and bounded subset (for example, the embedding of a finite connected graph on M). Let S be a connected component of $M - K$. Then there exists a surface $T \subseteq S$ such that ∂T consists of connected components $J_1, \cdots, J_s$, where each J_i is a simple closed curve. Moreover, the connected components of $S \setminus T$, denoted by $L_1, \cdots, L_s$, are all homeomorphic to hollow cylinders, each attached along one boundary component to J_i and along the other boundary component to a subset of K.

Using this lemma, instead of selecting the entire region S, we choose a large subsurface of it in such a way that capping this subsurface and turning it into a region homeomorphic to a disk guarantees that the entire region S also becomes homeomorphic to a disk. Repeating this process for every region that is not homeomorphic to a disk eventually produces a cellular embedding. This procedure is called the capping of the embedding ρ.

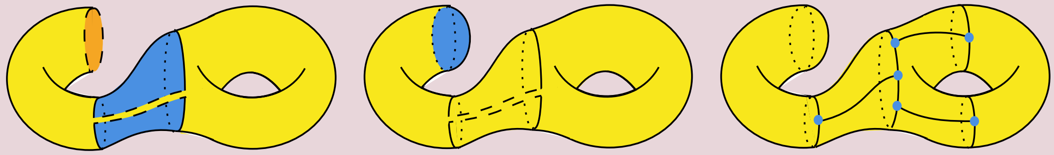
Step 1. To gain a better understanding of this theorem, let us work through an example step by step. Consider the following planar graph, embedded on the double torus as shown. One expects that the capping procedure will transform this embedding into an embedding on the sphere. We begin by considering the regions determined by the given embedding. Geometrically, this corresponds to cutting the surface along the edges of the given embedding, where each resulting piece is one of the connected components. (why?)



Step 2. After separating the cut pieces, we obtain the three pieces shown in the figure. The cutting curves are indicated by dashed lines, while hidden curves are shown by dotted lines. The inner side of the surface is distinguished by a darker (orange) color. Choose the region S as shown and select the subsurface T inside it. As can be seen, the boundary of T consists of two connected components, each of which is a simple closed curve. Moreover, $S \setminus T$ has one connected component, which is clearly homeomorphic to a hollow cylinder. Also note that one end of this cylinder is attached to J_1, while the other end is attached to an edge of the embedded graph. Following the procedure described in the definition, instead of removing the entire region S, we remove only the subsurface T from the original surface (the double torus), obtaining the figure on the right.

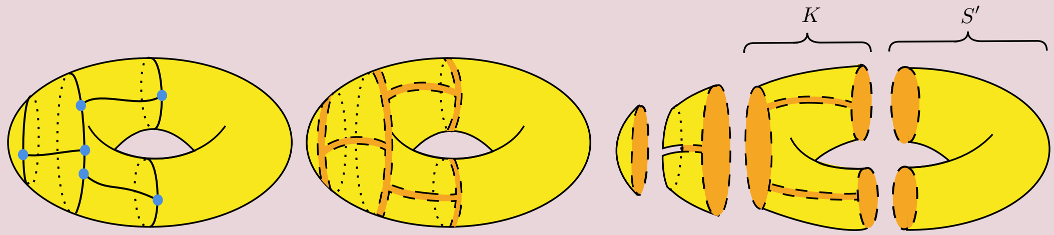


Step 3. The boundary of the surface obtained by removing the subsurface T consists of two connected components, each of which must be homeomorphic to a disk. (Why? Can you find a general argument for this statement?) Therefore, two disks may be attached along their boundaries to the connected components of the boundary of the resulting surface. (The disk currently being attached is shown in blue.) Through this process, the holes created in the original surface by removing T are sealed with "caps" in the form of disks. The figure on the right shows the embedding of the graph on the new surface, which is otherwise unchanged.



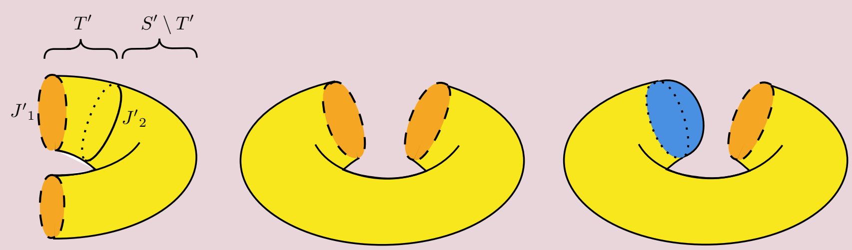
Problem 10. Explain why the condition that the connected components of $S \setminus T$ are homeomorphic to hollow cylinders is essential to the proof.

Step 4. With a little care, one sees that the surface obtained in the final figure of the previous step is in fact homeomorphic to an ordinary torus. Replacing the surface by this torus yields an embedding of the given graph on the torus, shown in the figure on the left. Thus, the capping procedure has been successful, decreasing the genus of the surface by one. We now repeat the first step by cutting the surface along the edges of the given embedding and separating the resulting pieces, obtaining the third figure. We must now choose a connected component of $M_1 \setminus \rho(G)$ that is not homeomorphic to a disk. We choose the region S' for the next stage of the procedure.

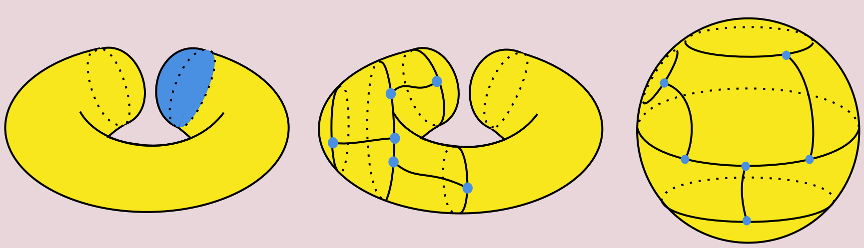


Problem 11. Is the region K homeomorphic to a disk? Argue to support your answer.

Step 5. According to the lemma, there exists a subsurface T' whose boundary components are all simple closed curves and such that the connected components of $M' \setminus T'$ are all homeomorphic to hollow cylinders. One such choice of T' is shown in the figure on the left. Removing this subsurface from M' (the entire surface) produces the middle figure. Finally, as before, each connected component of the boundary of the resulting surface is guaranteed to be homeomorphic to a disk, so each can be capped by attaching a disk. The attachment of a cap to one of the boundary components is shown.



Step 6. The figure on the left shows the capping of the remaining hole of the new surface. The middle figure shows the embedding of the given graph on the resulting surface (which is again the original embedding). Finally, observe that the surface shown in the middle figure is in fact the sphere. (why?) Thus, an embedding of the given graph on the sphere has been obtained, completing the procedure. The figure on the right shows the embedding of the graph on the sphere.



Problem 12. By repeating the capping procedure, show that eventually all regions become homeomorphic to disks, and consequently further capping can no longer alter the surface. At this stage, the capping procedure terminates, and the surface attains the maximum possible Euler characteristic.

Theorem 5. We state several properties of the capping operation. (justify these intuitively)

1. $\rho(G) \subset M^*$, and hence the embedding $\rho : G \to M$ naturally induces an embedding $\rho^* : G \to M^*$.
2. The embedding $\rho^* : G \to M^*$ is cellular, and moreover $|F_{\rho^*}| \geq |F_\rho|$.
3. The operation is independent of the order in which the connected components are chosen, and the surfaces obtained from different orders are homeomorphic.
4. If M is orientable, then M^* is also orientable. The converse, however, is false.
5. $M^* = M$ if and only if ρ is a cellular embedding. Otherwise, $\chi(M^*) > \chi(M)$. If M is orientable, then by property 4 the genus of M^* is also defined, and by theorem 3 we have $\gamma(M^*) < \gamma(M)$.

Definition 13. Let M be a connected closed surface. An embedding $\rho : G \to M$ is called a minimal embedding if for every connected closed surface M' and every embedding $\rho' : G \to M'$, we have $\chi(M) \geq \chi(M')$. If the surfaces M and M' are orientable, then by theorem 3 this is equivalently expressed as $\gamma(M) \leq \gamma(M')$.

Definition 14. Suppose that a finite connected graph G and a connected closed surface $M \subset \mathbb{R}^3$ are given. An embedding $\rho : G \to M$ is called a maximal embedding if for every connected closed surface M' and every embedding $\rho' : G \to M'$ of G on M', we have $|F_\rho| > |F_{\rho'}|$.

Theorem 6. A minimal embedding of a finite connected graph on a connected closed surface is cellular.

Proof. Suppose that the embedding $\rho : G \to M$ is minimal and, for the sake of contradiction, is not cellular. Then, by applying the capping operation to the surface M, Property 5 of theorem 6 guarantees the existence of a cellular embedding $\rho^* : G \to M^*$. However, $\chi(M^*) > \chi(M)$, contradicting the minimality assumption. □

Corollary 5. By theorem 5, for a minimal embedding $\rho : G \to M$ of a finite connected graph on a connected closed surface, the generalized Euler formula $|V| - |E| + |F_\rho| = \chi(M)$ holds.

Theorem 7. A minimal embedding of a finite connected graph on a connected closed surface is maximal.

Proof. Suppose that $\rho : G \to M$ is maximal and that $\rho' : G \to M'$ is minimal (and hence cellular). Then, by property 2 of theorem 5, the capped embedding $\rho^* : G \to M^*$ is maximal and cellular. Therefore,

$$|V| - |E| + |F_{\rho'}| \overset{\text{Euler identity}}{=} \chi(M') \overset{\text{Minimality}}{\geq} \chi(M^*) \overset{\text{Euler identity}}{=} |V| - |E| + |F_{\rho^*}| \overset{\text{Maximality}}{\geq} |V| - |E| + |F_{\rho'}|$$

$$\implies |V| - |E| + |F_{\rho^*}| = |V| - |E| + |F_{\rho'}| \implies |F_\rho| = |F_{\rho^*}| = |F_{\rho'}| \implies \rho' \text{ is a maximal embedding} \qquad \square$$

This former theorem is one of the most important theorems of this article.

Problem 13. Prove that every maximal cellular embedding is also a minimal embedding. Also show that both of these conditions are necessary by constructing appropriate counterexamples.

Definition 15. Let M be a connected closed orientable surface. An embedding $\rho : G \to M$ is called a minimal orientable embedding if for every connected closed orientable surface M' and every embedding $\rho' : G \to M'$, we have $\chi(M) \geq \chi(M')$. By theorem 3, this is equivalently expressed as $\gamma(M) \leq \gamma(M')$.

Definition 16. Suppose that a finite connected graph G and a connected closed orientable surface $M \subset \mathbb{R}^3$ are given. An embedding $\rho : G \to M$ is called a maximal orientable embedding if for every connected closed orientable surface M' and every embedding $\rho' : G \to M'$, we have $|F_\rho| > |F_{\rho'}|$.

Analogously to theorem 6 and theorem 7, together with the result of problem 13, the following theorem can also be proved using theorem 5 to extend the Euler-type formulae to embedded graphs.

Theorem 8. An embedding $\rho : G \to M$ of a finite connected graph G on a connected closed orientable surface M is a minimal orientable embedding if and only if it is maximal orientable and cellular. Consequently, the generalized Euler formula (theorem 4) holds for minimal orientable embeddings.

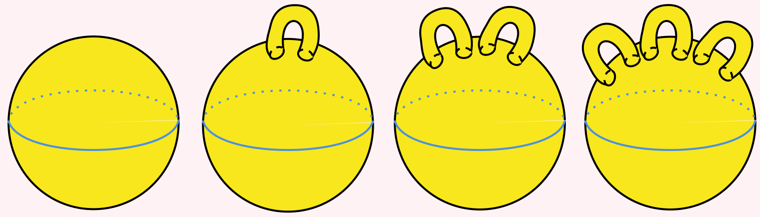
This implies that in the special case where the embedding map is either minimal or minimal orientable, the number of regions is determined by theorem 4 and is a graph invariant.

Theorem 9. The number of regions determined by a finite connected graph G on a connected closed surface M under a minimal or minimal orientable embedding such as $\rho : G \to M$ is independent of the embedding map. Consequently, the quantity $|F_\rho|$ is both a graph invariant and a topological invariant.

Definition 17. For every finite connected graph G, the smallest integer $g \in \mathbb{Z}^{\geq 0}$ such that an embedding of G into M_g exists is called the genus of the graph G, and is denoted by $\gamma(G)$. Note that one must first prove the existence of an integer $g \in \mathbb{Z}^{\geq 0}$ for which an embedding of G into M_g exists.

Remark 4. As you have seen, definition 13 is restricted to embeddings on orientable surfaces. It is therefore natural to extend this concept to arbitrary surfaces by using the Euler characteristic instead of the genus. We define the surface number of a graph G to be the largest integer $n \in \mathbb{Z}$ such that there exists an embedding $\rho : G \to M$ with $\chi(M) = 2 - 2n$. The surface number of G is denoted by $\mu(G)$. Observe that, by definition, for every graph G, the quantity $2 - 2\gamma(G)$ is the largest possible Euler characteristic of an orientable surface admitting an embedding of G, whereas $2 - 2\mu(G)$ is the largest possible Euler characteristic of an arbitrary surface M admitting an embedding of G. A natural question is whether the surface number and the genus coincide for every graph G. Equivalently, does every minimal embedding $\rho : G \to M$ necessarily admit an embedding $\rho' : G \to M'$ such that $\chi(M') = \chi(M)$ and M' is orientable? We postpone the answer to this question until the end of this article.

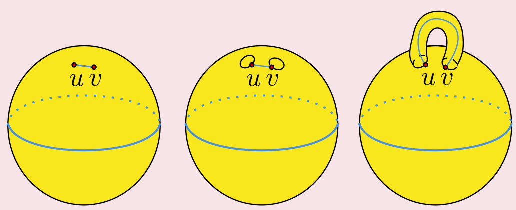
Definition 18. The surface obtained by attaching n hollow cylinders to the surface of a sphere (as shown below) is called the n-handle sphere.



Problem 14. Prove that the n-handle sphere is homeomorphic to the torus with n holes (M_g).

Theorem 10. Every finite connected graph has a genus. In other words, there exists an integer $g \in \mathbb{Z}^{\geq 0}$ such that an embedding of G into M_g exists.

Proof. Place the vertices of G on the sphere. For each edge $\overline{uv} = e \in E(G)$, choose two sufficiently small disks containing u and v, respectively, and attach the boundary components of a hollow cylinder to these two disks. Denote this handle by $H_{\overline{uv}}$. On the resulting surface ($M_{|E(G)|}$), route each edge $\overline{uv} = e \in E(G)$ through the corresponding handle $H_{\overline{uv}}$. It is clear that this construction yields an embedding of G into $M_{|E(G)|}$. □



Corollary 6. The genus of every finite connected graph is well defined. Let $\rho : G \to M$ be a minimal orientable embedding. By theorem 8, this embedding is also cellular. Hence, by theorem 3 and theorem 4, together with definition 15, we obtain $|V| - |E| + |F_\rho| = \chi(M) = 2 - 2\gamma(M) = 2 - 2\gamma(G)$.

Theorem 11. For a finite connected graph G, a connected closed surface M, and an embedding $\rho : G \to M$, we have $|V| - |E| + |F_\rho| \geq \chi(M)$.

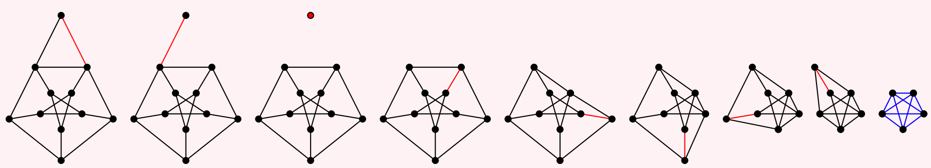
Proof. First, note that by the definition of a surface, the embedding ρ can be extended to a cellular embedding $\tilde{\rho} : \tilde{G} \to M$. (why?) That is, there exists a finite connected graph $\tilde{G}$ such that G is a subgraph of $\tilde{G}$ and $\tilde{\rho}\big|_G \equiv \rho(G)$. By the generalized Euler formula, we then have $|V(\tilde{G})| - |E(\tilde{G})| + |F_{\tilde{\rho}}| = \chi(M)$. Now consider, one by one, all edges $e \in E(\tilde{G}) \setminus E(G)$. If e lies on the boundary of two distinct regions, then deleting e from the embedded graph $\tilde{G}$ decreases the number of regions ($|F_{\tilde{\rho}}|$) by exactly one. Consequently, the quantity $|V(\tilde{G})| - |E(\tilde{G})| + |F_{\tilde{\rho}}|$ remains unchanged. If, on the other hand, e lies on the boundary of only one region, then there exists a vertex v such that $\deg(v) = 1$. (why?) Deleting v together with its incident edge again leaves the quantity $|V(\tilde{G})| - |E(\tilde{G})| + |F_{\tilde{\rho}}|$ unchanged. Thus, all edges in $E(\tilde{G}) \setminus E(G)$ can be removed without changing this quantity. At this stage, every vertex in $V(\tilde{G}) \setminus V(G)$ has degree zero. (why?) Removing these vertices does not change $|F_{\tilde{\rho}}|$. (why?) Therefore, the quantity $|V(\tilde{G})| - |E(\tilde{G})| + |F_{\tilde{\rho}}|$ decreases. Completing this process finishes the proof. □

When investigating whether a graph can be embedded on a surface, Euler's formula is not particularly effective because the number of regions is generally unknown. As in Corollaries 2 and 3, we therefore seek a criterion involving only the graph and the surface. The following problem provides such a criterion.

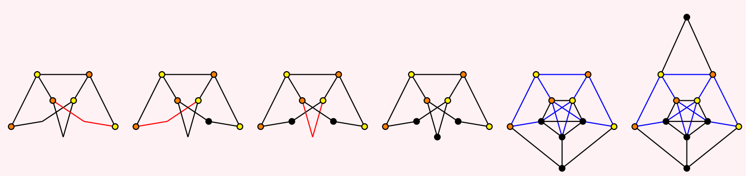
Problem 15. Suppose that G is a finite connected graph containing no cycle of length less than k. Then for every connected closed orientable surface M and every embedding $\rho : G \to M$, we have $k|V| + 2k\gamma(M) \geq 2k + (k-2)|E|$. Show that $K_{5,5}$ cannot be embedded on the torus. Also formulate precisely the equality case of this inequality for every positive integer k.

Definition 19. An edge contraction is the operation of deleting an edge and identifying its two endpoints, while an edge subdivision is the operation of inserting a new vertex on an edge. A graph obtained from G by a sequence of edge contractions, vertex deletions, and edge deletions is called a minor of G. A minor of G that is isomorphic to a graph F is called an F-minor of G. A graph obtained from G by a sequence of edge subdivisions is called a subdivision of G. We call a graph G an F-subdivision if it contains a subgraph isomorphic to a subdivision of F.

Example 8. The construction of a K_5-minor of the Horned Petersen graph is illustrated in the figure. A red vertex indicates a vertex deletion, and a red edge indicates an edge contraction. A subgraph is highlighted in blue.



Example 9. The construction of the Horned Petersen graph as a $K_{3,3}$-subdivision is illustrated in the figure. The first graph is $K_{3,3}$, whose vertices are distinguished by two colors. A red edge indicates that the edge is subdivided, while the blue coloring has the same meaning as in the previous example.



Problem 16. Prove that the Horned Petersen graph is not a K_5-subdivision.

Problem 17. If G is an F-subdivision, then it contains an F-minor. The converse is not necessarily true.

We now state two well-known theorems characterizing planar graphs in terms of minors and subdivisions.

Theorem 12. (**Kuratowski**)[5]: A connected graph G is planar if and only if it contains neither a K_5-subdivision nor a $K_{3,3}$-subdivision.

Theorem 13. (**Wagner**)[9]: A connected graph is planar if and only if it contains no a K_5 or $K_{3,3}$ minor.

Corollary 7. A connected graph contains either a K_5-subdivision or a $K_{3,3}$-subdivision if and only if it contains either a K_5-minor or a $K_{3,3}$-minor. Can you prove this fact without using the two theorems above?

Does an analogous theorem exist for surfaces other than the plane? The following theorem provides an answer.

Definition 20. A family $\mathcal{F}$ of graphs is minor-closed if any minor of $G \in \mathcal{F}$ belongs to $\mathcal{F}$.

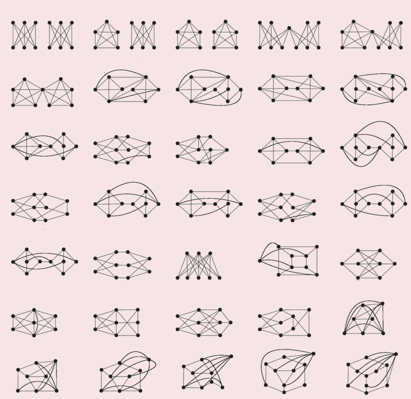
Remark 5. For every connected closed surface M, the family of graphs embeddable on M is minor-closed.

The following theorem is a consequence of one of the most fundamental results in graph minor theory.

Theorem 14. Robertson–Seymour theorem:[8] If a family of graphs $\mathcal{F}$ is minor-closed, then there exists a finite set S of graphs such that, for every graph G, we have $G \in \mathcal{F}$ if and only if, for every graph $H \in S$, the graph G contains no H-minor. The set S is called the set of forbidden minors of the family $\mathcal{F}$. By the previous remark, this theorem applies to the family of graphs embeddable on a given surface.

Example 10. Glover, Huneke, and Wang proposed a conjectural list of forbidden minors for the Möbius band.[3] Although they did not prove the result, Archdeacon[1] and Arash Asadi Shahmirzadi[2] proved it independently.

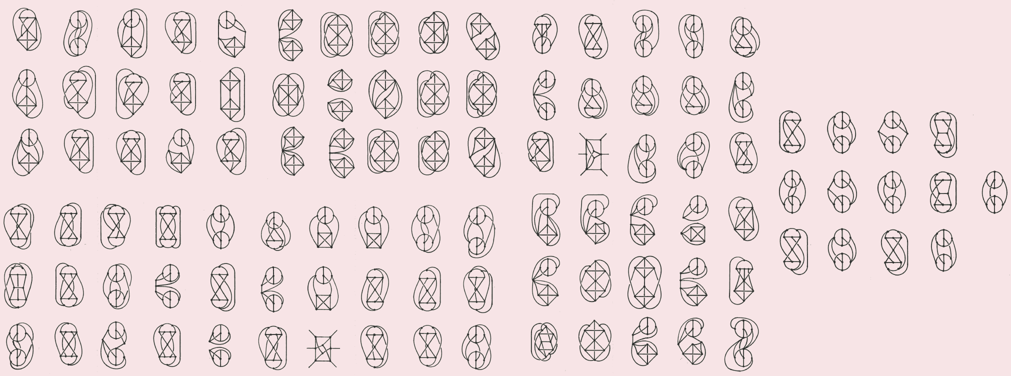
Theorem 15. There is a list J of 35 irreducible forbidden graphs for the Möbius band such that a graph G is embeddable on the Möbius band if and only if, for every graph $H \in J$, G contains no H-minor.



The corresponding theorem does not hold for subdivisions. Explaining the reason requires a complete discussion of the original statement of the theorem, together with the notions of categorical order and categorical antichains, which is beyond the scope of this article. (For further reading, see this article.) Moreover, one can say even more: a theorem analogous to corollary 7 also fails to hold. A celebrated result of Glover, Huneke, and Wang classifies the family of graphs embeddable on the Möbius band (shown in problem 9) in terms of subdivisions as follows:

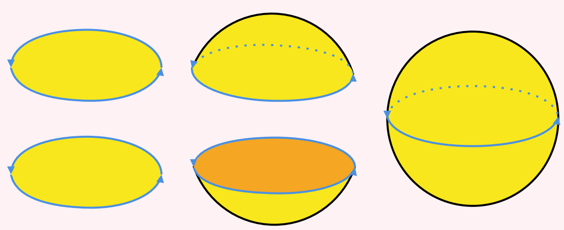
Definition 21. A finite connected graph G is called an irreducible forbidden graph for a connected closed surface M if it is not embeddable on M, but every proper subgraph of G is embeddable on M. For example, $K_{3,3}$ and K_5 are both irreducible forbidden graphs for the plane.

Theorem 16. Glover-Huneke-Wang Theorem:[3] There exists a list I of 103 irreducible forbidden graphs for the Möbius band such that a graph G is embeddable on the Möbius band if and only if G contains no H-subdivision for any graph $H \in I$.



We now return to remark 4. We first need to define the projective plane. Observe first that the boundary of the Möbius band is a closed curve and is therefore homeomorphic to a circle. (why?) Hence, one may assign an orientation to the boundary of the Möbius band. Likewise, the boundary of a closed disk is clearly a circle and can therefore also be oriented. It is helpful to begin with the following warm-up example:

Example 11. The sphere is the surface obtained by gluing together two warped disks along their oriented boundaries according to the chosen orientations.



The projective plane is the surface obtained by gluing a Möbius band and a closed disk together along their boundaries according to the chosen orientations. If this construction is difficult to visualize, it is probably because, in order to realize it in $\mathbb{R}^3$, the interior of the Möbius band and the interior of the disk must intersect. Consequently, the projective plane cannot be embedded in three-dimensional space. However, it can be shown that it is embeddable in four-dimensional space.

Problem 18. Prove that a finite graph is embeddable on the projective plane if and only if it is embeddable on the Möbius band. Use the same strategy as in problem 2.

Problem 19. Construct a CW-complex for the Möbius band, and hence for the projective plane. Prove that the Euler characteristic of the projective plane is 1, whereas the Euler characteristic of the Möbius band is 0.

Problem 20. Prove that the graph K_5 is embeddable on the Möbius band, and hence on the projective plane. Therefore, $\gamma(K_5) = 2,\ \mu(K_5) = 0$, which gives a negative answer to remark 4.

We finally return to Kuratowski's theorem and ask a question analogous to the one posed at the beginning of this article. Why are K_5 and $K_{3,3}$ forbidden graphs for the plane? The answer lies in the following theorems, which determine the genera of several well-known graph families. We state these results without proof.

Theorem 17. Ringel: $\gamma(K_{m,n}) = \left\lceil \frac{(m-2)(n-2)}{4} \right\rceil,\ \forall m, n \geq 2.$

Proof. To prove this theorem, we first establish the following weaker statement.

Lemma 2. $\gamma(K_{m,n}) \geq \left\lceil \frac{(m-2)(n-2)}{4} \right\rceil,\ \forall m, n \geq 2.$

Proof. Since $K_{m,n}$ is bipartite, every cycle has length at least 4. As every edge appears exactly twice in the boundary traversal of the regions of any cellular embedding, we have $2|E| \geq 4|F_\rho|$, where $\rho : K_{m,n} \to M$ is an arbitrary embedding. Moreover, $|E(K_{m,n})| = mn$ and $|V(K_{m,n})| = m+n$, so $\frac{mn}{2} \geq |F_\rho|$. However, the generalized Euler formula gives $|V| - |E| + |F_\rho| = 2 - 2\gamma$ for a minimal embedding ρ. Hence,

$$(m+n) - mn + \frac{mn}{2} \geq |V| - |E| + |F_\rho| = 2 - 2\gamma \iff 2\gamma \geq \frac{mn}{2} - m - n + 2.$$

Therefore,

$$\gamma \geq \frac{mn - 2m - 2n + 4}{4} = \frac{(m-2)(n-2)}{4}.$$

□

Now suppose that both m and n are even. Label the vertex sets by $\{0, 2, \cdots, m-2\}$ and $\{1, 3, 5, \cdots, n-1\}$. Consider the following embedding preserving the cyclic order:

$$\begin{bmatrix} 0(\mathrm{mod}4) & 1 & 3 & 5 & \cdots & n-3 & n-1 \\ 2(\mathrm{mod}4) & n-1 & n-3 & n-5 & \cdots & 3 & 1 \\ 1(\mathrm{mod}4) & 0 & 2 & 4 & \cdots & m-4 & m-2 \\ 3(\mathrm{mod}4) & m-2 & m-4 & m-6 & \cdots & 2 & 0 \end{bmatrix}$$

All regions of this embedding are quadrilaterals. Consequently, equality holds in the lemma, proving the desired result. The remaining parity cases are established inductively. □

Theorem 18. Ringel-Youngs: $\gamma(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil,\ \forall n \geq 3.$

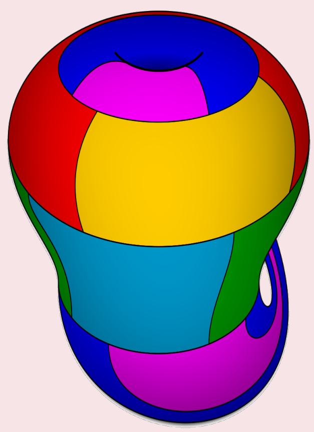
Proof. If $\rho : K_n \to M$ is a minimal embedding, then $|V| - |E| + |F_\rho| = 2 - 2\gamma$. Since every cycle has at least three edges and every edge appears at least twice in the boundary traversal of the regions, we have $2|E| \geq 3|F_\rho|$. Proceeding as in the previous theorem $|V| - |E| + \frac{2}{3}|F_\rho| \geq 2 - 2\gamma$ and therefore

$$2\gamma \geq 2 - n + \frac{n(n-1)}{6} = \frac{n^2 - 7n + 12}{6} \implies \gamma \geq \frac{(n-3)(n-4)}{12}$$ □

Theorem 19. White: $\gamma(K_{m,n,mn}) = \frac{(mn-2)(n-1)}{2}$, $\forall m,n$, where $K_{m,n,p}$ denotes the complete tripartite graph.

In fact, there is a profound relationship between the chromatic number of a graph and its genus.

Proposition 1. Ringel-Youngs:[6] For every finite graph G, exactly $\left\lfloor \frac{7+\sqrt{1+48\gamma(G)}}{2} \right\rfloor$ colors are sufficient to color its vertices. This theorem shows that the number of colors required to color any finite connected graph G admitting a cellular embedding on a surface M is always the same. Consequently, one may define an analogue of the chromatic number for a surface, denoted by $\chi_C(M)$. The theorem states that for every orientable surface M, $\chi_C(M) = \left\lfloor \frac{7+\sqrt{1+48\gamma(M)}}{2} \right\rfloor$. As a generalization of this result, Ringel and Youngs also proved that, after extending the statement to non-orientable surfaces (and naturally replacing the genus by the Euler characteristic), the formula $\chi_C(M) = \left\lfloor \frac{7+\sqrt{49-24\chi(M)}}{2} \right\rfloor$ holds for every connected closed surface except for exactly one non-orientable surface, namely the Klein bottle! The Euler characteristic of the Klein bottle is 0, so this theorem predicts that its chromatic number should be 7. However, the figure below shows a coloring of the Klein bottle using only 6 colors.



Proof. Note that it suffices to prove the statement for critical graphs. A graph is called critical if every nontrivial subgraph has strictly smaller chromatic number. By a well-known theorem, $\chi_C(G) - 1 \leq \frac{2|E|}{|V|}$. Let $\rho : G \to M$ be a minimal embedding. Then we have $|V| - |E| + |F_\rho| = 2 - 2\gamma$, or equivalently $|E| \leq |V| + |F_\rho| - (2 - 2\gamma)$. The length of the shortest cycle in a graph is 3, and each edge appears exactly twice in the boundary walks of the regions. Hence $2|E| \geq 3|F_\rho|$, and therefore:

$$3|E| \leq 3|F_\rho| + 3|V| - 3(2-2\gamma) \leq 2|E| + 3|V| - 3(2-2\gamma) \iff |E| \leq 3|V| - 6 + 6\gamma$$

Thus, $2|E| \leq 6|V| - 12 + 12\gamma$. Combining this with the theorem stated at the beginning of the proof yields $(\chi_C(G) - 1)|V| \leq 6|V| - 12 + 12\gamma$. If $\gamma \geq 1$, then since $|V| \geq \chi_C(G)$, we obtain $\chi_C(G) - 1 \leq 6 + \frac{12\gamma - 12}{\chi_C(G)} \iff \chi_C(G)^2 - 7\chi_C(G) - (12\gamma - 12) \leq 0$. Solving this quadratic inequality gives:

$$\left(\chi_C(G) - \frac{7+\sqrt{1+48\gamma}}{2}\right)\left(\chi_C(G) - \frac{7-\sqrt{1+48\gamma}}{2}\right) \leq 0.$$

Since $\gamma \geq 1$, we have $\sqrt{1+48\gamma} \geq 7$, and hence $\frac{7-\sqrt{1+48\gamma}}{2} \leq 0$. Since $\chi_C(G) \geq 1$, the second factor is positive, and therefore the first factor must be nonpositive, proving one direction of the theorem. $\square$

In the case of disconnected graphs, a remarkable theorem of Battle, Harary, Kodama, and Youngs states the following.

Theorem 20. If a graph G has connected components $G_1, \cdots, G_k$, then $\gamma(G) = \sum_{i=1}^{k} \gamma(G_i)$.

Problem 21. Prove that if a finite graph G is embeddable on the torus, then its vertices can be colored with at most 7 colors so that adjacent vertices receive different colors. (A much deeper analogous result states that every planar graph is 4-colorable.)

Problem 22. (IMO 1986): An integer is assigned to each vertex of a regular pentagon so that the sum of the five integers is positive. If there exist three consecutive vertices with integers (in order) x, y, z, where $y < 0$, then these three integers may be replaced by $x+y, -y, z+y$, respectively. Does this process terminate for every initial assignment of integers to the vertices?

We now state a stronger version of this problem, which generalizes the IMO problem to arbitrary graphs.

Definition 22. A finite connected graph G with an integer assigned to each of its vertices is called a graded graph. We define two operations on a graded graph. The operation of borrowing at a vertex v decreases the integer at each neighbor of v by one and adds one to the integer at v. Similarly, the operation of lending at a vertex v increases the integer at each neighbor of v by one while decreasing the integer at v by one. A vertex is called indebted if the integer assigned to it is negative. A graded graph is called communistic! if, by a sequence of borrowing and lending operations, it is possible to eliminate all indebted vertices.

Example 12. The previous problem states that the graded graph C_5 is communistic provided that the sum of the integers assigned to its vertices is positive. Can you find a graded graph on C_5 with total sum zero that is not communistic?

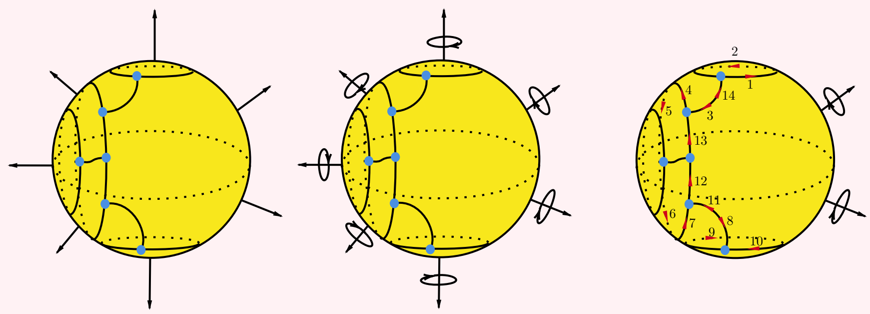
Problem 23. Prove that if G is a communistic graded graph, then all indebted vertices can be eliminated using only borrowing operations. This shows that the next problem is indeed a generalization of the IMO problem.

Problem 24. Prove that a graded graph is communistic if the sum of the integers assigned to its vertices is at least $|E| - |V| + 1$. This quantity is called the Betti number of the graph, denoted by $\beta(G)$, and, similarly to the genus, it is a topological invariant of the graph.

Finally, we conclude this article with an exceptionally amusing problem!

Remark 6. Suppose that $\rho : G \to M$ is a cellular embedding of a finite connected graph on a connected closed orientable surface. Then every connected component of $M \setminus \rho(G)$ is homeomorphic to a disk. Since the surface M is orientable, it induces an orientation on each of these connected components. Because each of these components is homeomorphic to a two-dimensional disk, the chosen orientation of M induces an orientation on each disk. Note that under the homeomorphism between each region determined by the graph on the surface and the disk, the boundary of the region is homeomorphic to the boundary of the disk. Now observe that choosing an orientation on a disk determines a counterclockwise orientation of its boundary, which in turn induces a counterclockwise orientation on the boundary of each region determined by the graph on the surface.

Example 13. An example of the orientation induced on the final embedding in the capping example (the final figure of Step 6) by the natural outward orientation of the sphere is shown below. The first figure shows the chosen normal vectors at various points of the sphere. The second figure shows the counterclockwise orientation induced around each normal vector. In the third figure, the induced orientation on the boundary of one of the regions of the sphere is indicated by the red arrows. The numbers next to the arrows specify the order in which the boundary of the region is traversed.



Problem 25. Prove that every edge is oriented exactly twice, either by two distinct regions or twice by the same region, and that these two orientations are opposite.

Now you are ready to think about these amusing problems.

Problem 26. Suppose that $\rho : G \to M$ is a cellular embedding of a finite connected graph on a connected closed orientable surface. For each connected component of $M \setminus \rho(G)$, place one car on the boundary of that region (which is homeomorphic to a two-dimensional disk). Each car moves continuously in the counterclockwise direction along the boundary of its corresponding region.

1. **(St. Petersburg 1993)**: Suppose that M is the unit sphere in $\mathbb{R}^3$, and that every car moves counterclockwise along the boundary of its region with speed at least 1 mm/h. Note that the motion of the cars is completely arbitrary: it need not be periodic, it need not have constant speed, and it may involve both positive and negative acceleration. Prove that a collision must occur in finite time.
2. Instead of assuming a positive lower bound on the speed, suppose that each car completes every traversal of the boundary of its region in finite time. Note that this assumption is stronger than the previous one, since it allows cars to stop during their motion. Prove that collisions must occur at at least two distinct points of the surface in finite time.[4] Is this a geometric property of the graph embedding?
3. Find an example of cars moving on a torus in such a way that no collision ever occurs.
4. Now suppose that there are d_F cars moving along the boundary of each connected component $F \subseteq M \setminus \rho(G)$, where $d_F \in \mathbb{N}$. Furthermore, suppose that the boundary of each region F is partitioned into d_F connected arcs in such a way that at every moment at most one car occupies each arc. Note that stopping is still allowed under these assumptions. Prove that if M is the sphere, then at least $2 + \sum(d_F - 1)$ collisions occur, where the sum is taken over all connected components of $M \setminus \rho(G)$.
5. If M is the torus and at least one region contains more than one car, then a collision must occur.
6. **Klyachko**: For an arbitrary surface M, collisions occur in at least $\chi(G) + \sum(d_F - 1)$ distinct points.

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